THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The LCM of 30, 42 and 66 is: 1

(a) 6930 **(b)** 1155 **(c)** 2310 **(d)** 1260

2. If α, β are zeroes of $x^2 - 4x + 3$, then $\alpha^3 + \beta^3$ is: 1

(a) 28 **(b)** 100 **(c)** 40 **(d)** 64

3. The pair $kx + 2y = k - 2$ and $8x + ky = k$ has no solution when $k =$ 1

(a) 4 **(b)** -4 **(c)** 2 **(d)** 8

4. If the roots of $x^2 - 7x + 10 = 0$ are α, β, then $\alpha^2 + \beta^2 =$ 1

(a) 49 **(b)** 29 **(c)** 39 **(d)** 69

5. If the 4th term of an AP is 9 and the 9th term is 4, then the 13th term is: 1

(a) 13 **(b)** 36 **(c)** 1 **(d)** 0

6. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 1 : 2$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 1: 1

(a) 10 **(b)** 4 **(c)** 9 **(d)** 8

7. The point on the y-axis equidistant from $(2, 3)$ and $(6, 5)$ has $y =$ 1

(a) 12 **(b)** 4 **(c)** 24 **(d)** -12

8. If $\sin\theta = \frac{4}{5}$ then $\sin^2\theta + \cos^2\theta + \tan^2\theta =$ 1

(a) $\frac{16}{9}$ **(b)** $\frac{25}{9}$ **(c)** $\frac{16}{25}$ **(d)** $\frac{9}{25}$

9. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 4 m towards the foot of the tower along a straight line, the angle of elevation becomes 45°. The height of the tower is: 1

(a) $6 + 2\sqrt{3}$ m **(b)** 4 m **(c)** $2 + 2\sqrt{3}$ m **(d)** $\frac{4\sqrt{3}}{3}$ m

10. Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 30 cm. The length of each tangent is 40 cm. The length of the chord AB is: 1

(a) 24 cm **(b)** 80 cm **(c)** 60 cm **(d)** 48 cm

11. A chord of a circle of radius 7 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) The area of the minor segment is: 1

(a) 25.67 cm² **(b)** 46.86 cm² **(c)** 4.47 cm² **(d)** 21.19 cm²

12. The largest possible solid sphere is carved out of a cube of edge 21 cm. The volume of the material wasted is: 1

(a) 4410 cm³ **(b)** 9261 cm³ **(c)** 6835.5 cm³ **(d)** 4851 cm³

13. The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70, 70–80 and frequencies 3, 4, p, 4, 15, 10, 19, 22 respectively is 49. The value of p is: 1

(a) 25 **(b)** 28 **(c)** 22 **(d)** 23

14. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum of the numbers is 8 is: 1

(a) $\frac{1}{9}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{31}{36}$ **(d)** $\frac{5}{36}$

15. The least number that is divisible by every number from 1 to 6 is: 1

(a) 30 **(b)** 60 **(c)** 120 **(d)** 360

16. If α, β are zeroes of $x^2 - 5x + 6$, then $\alpha^2\beta + \alpha\beta^2$ is: 1

(a) 36 **(b)** 30 **(c)** 25 **(d)** 11

17. **Assertion (A):** The HCF of 12 and 18 is 6 and their LCM is 36. 1
Reason (R): For two positive integers a and b, HCF(a, b) x LCM(a, b) = a x b.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** A quadratic polynomial whose graph is a parabola cutting the x-axis at two distinct points has two real zeros. 1
Reason (R): The zeros of a polynomial $p(x)$ are the x-coordinates of the points where the graph of $y = p(x)$ meets the x-axis.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** The graphs of $x + y = 4$ and $x - y = 2$ intersect at the point $(3, 1)$. 1
Reason (R): The coordinates of the point of intersection of the lines of a pair of linear equations give the solution of the pair.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

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20. **Assertion (A):** The equation $2x^2 - 3x + 5 = 0$ has no real roots. 1

Reason (R): If the discriminant $b^2 - 4ac < 0$, a quadratic equation has no real roots.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Two numbers have HCF 12 and are in the ratio $3 : 5$. Find the numbers and their LCM. 2

22. If α, β are zeroes of $x^2 - 4x + 3$, form a quadratic polynomial whose zeroes are α^2 and β^2. 2

23. Find k so that the pair $kx + 2y = k - 2$, $8x + ky = k$ has no solution. 2

24. Find the values of k for which $kx^2 + 4x + 1 = 0$ has real and equal roots. 2

25. The 4th term of an AP is 9 and the 9th term is 4. Find the 13th term. 2

26. In $\triangle ABC$, $DE \parallel BC$, $AD = 3$ cm, $AB = 7$ cm and $BC = 14$ cm. Find DE. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. A, B and C take 18, 24 and 36 minutes per round of a circular track. They start together. After how many minutes do they meet at the start again, and how many rounds has each completed? 3

28. Find a quadratic polynomial whose zeroes are $\alpha + \dfrac{1}{\beta}$ and $\beta + \dfrac{1}{\alpha}$, where α, β are zeroes of $x^2 - 3x + 2$. 3

29. Solve: $\dfrac{2}{x} + \dfrac{3}{y} = 2$ and $\dfrac{3}{x} + \dfrac{5}{y} = \frac{19}{6}$. 3

30. Solve for x: $\dfrac{x+1}{x-1} + \dfrac{x-1}{x+1} = \frac{10}{3}$. 3

31. In an AP, $S_6 = 75$ and $S_{10} = 185$. Find the AP. 3

OR

In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 8$ cm and $DC = 12$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. 3

32. Show that the points $A(1, 1)$, $B(5, 4)$, $C(10, 4)$, $D(6, 1)$ are the vertices of a rhombus. 3

OR

Prove that: $\tan\theta + \cot\theta = \sec\theta \csc\theta$. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** A school is arranging chairs for the annual day. There are 144 chairs in the front block and 180 chairs in the back block. The teacher wants to arrange each block in rows so that every row has the same number of chairs, and every row in both blocks has equal length. 5

(i) Express 144 and 180 as products of prime factors.

(ii) Find the greatest possible number of chairs in each row.

(iii) How many rows will there be in all?

(iv) The coordinator proposes one more block of 216 chairs with the same rule. Find the greatest row length that works for all three blocks and the total number of rows.

34. **Case study.** Meera is designing a parabola-shaped flower bed in her school garden. She models its boundary by the graph of $y = p(x) = x^2 - 8x + 15$, where x is measured in metres along the ground. 5

(i) Find the zeros of $p(x)$.

(ii) Find the sum and the product of the zeros.

(iii) How many times does the graph of $y = p(x)$ meet the x-axis, and at which points?

(iv) Form a quadratic polynomial whose zeros are the reciprocals of the zeros of $p(x)$.

35. A boat goes 36 km upstream and 40 km downstream in 10 hours, and 24 km upstream and 70 km downstream in 11 hours. Find the speed of the boat in still water and the speed of the stream. 5

36. Two taps running together fill a tank in $\frac{10}{3}$ hours. The larger tap alone takes 5 hours less than the smaller one. Find the time each takes alone. 5

OR

A ladder has 21 rungs whose lengths decrease uniformly from 100 cm at the bottom to 40 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

37. Two ships leave a port at the same time; one sails 60 km due north and the other 80 km due west. Find the distance between them and the ratio of this distance to the one if both had sailed only half as far. 5

OR

The vertices of $\triangle ABC$ are $A(0,\,0)$, $B(12,\,0)$ and $C(6,\,8)$. Find the coordinates of the centroid and the area of the triangle. 5

38. Prove that: $(\csc\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta) = 1$. 5

OR

A building is 4 m high. From the top of the building the angle of elevation of the top of a tower is 45° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). 5

END OF QUESTION PAPER — Total: 80 Marks

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